

# Tests.pdf - ECE 422C Assignment 2

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**Assignment:** Mastermind - Phase I + Wordle - Phase II

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## Test 1 - Invalid Game Name

**Purpose:** Verify fall-back mechanism and make sure that the game loads only with a proper input.

**Input:**

java assignment2.Driver master  
+ java assignment2.Driver wordie

**Expected Output:**

java.lang.IllegalArgumentException: Unknown game: master  
+ java.lang.IllegalArgumentException: Unknown game: wordie

**Actual Output:**

java.lang.IllegalArgumentException: Unknown game: master < **Correct!**  
+ java.lang.IllegalArgumentException: Unknown game: wordie < **Correct!**

## Test 2 - Invalid Input

**Purpose:** Verify that the program rejects guesses with an (1) incorrect number of pegs and guesses containing (2) illegal colors.

**Input #1:** Please enter your guess: **BGO**

**Input #2** Please enter your guess: **BGXZ**

**Expected Output #1:** Please Try Again. Reason: Guess must contain exactly 4 colors.

**Expected Output #2:** Please Try Again. Reason: Input color {X} is not a legal color for this game configuration.

**Actual Output #1:** Please Try Again. Reason: Guess must contain exactly 4 colors. < **Correct!**

**Actual Output #2:** Please Try Again. Reason: Input color {X} is not a legal color for this game configuration. < **Correct!**

## Test 2 - Invalid Input

**Purpose:** Verify that the program rejects guesses with an (1) incorrect number of letters and guesses containing (2) illegal characters, and (3) guesses not in the dictionary.

**Input #1:** Please enter your guess: **SINCERELY**

**Input #2:** Please enter your guess: **HE'LL**

**Input #3:** Please enter your guess: **MESSI**

**Expected Output #1:** Please Try Again. Reason: Guess must contain exactly 5 letters.

**Expected Output #2:** Please Try Again. Reason: {} is not a letter.  
**Expected Output #3:** Please Try Again. Reason: {MESSI} is not a word in the dictionary that was provided.

**Actual Output #1:** Please Try Again. Reason: Guess must contain exactly 5 letters. < **Correct!**  
**Actual Output #2:** Please Try Again. Reason: {} is not a letter. < **Correct!**  
**Actual Output #3:** Please Try Again. Reason: {MESSI} is not a word in the dictionary that was provided. < **Correct!**

### Test 3 - Duplicate Colors Without Exact Matches

**Purpose:** Verify that the program correctly handles repeated colors when there are no exact matches and counts each peg at most once.

**Input:** Please enter your guess: **RRBG** (with Secret Code as: *BGOR*)

**Expected Output:** 0B\_3W

**Actual Output:** 0B\_3W < **Correct!**

### Test 3 - Duplicate Letters Without Exact Matches

**Purpose:** Verify that the program correctly handles repeated letters when there are no exact matches and counts each result category at most once.

**Input:** Please enter your guess: **SHEEP** (with Secret Code as: *HOUSE*)

**Expected Output:** P P P A A

**Actual Output:** P P P A A < **Correct!**

### Test 4 - Duplicate Colors Exact Matches

**Purpose:** Verify that exact matches are assigned first, and that once a secret color in the code has been used by a black peg, the same color elsewhere in the guess can't also earn a white.

**Input:** Please enter your guess: **BBBB** (with Secret Code as: *PYOB*)

**Expected Output:** 1B\_0W

**Actual Output:** 1B\_0W < **Correct!**

### Test 4 - Duplicate Letters Exact Matches

**Purpose:** Verify that exact matches are assigned first, and that once a secret letter in the code has been used by a 'C' flag, the same color elsewhere in the guess can't also earn a 'C'.

**Input:** Please enter your guess: **SMALL** (with Secret Code as: *APPLE*)

**Expected Output:** A A P C A

**Actual Output:** A A P C A < Correct!

### Test 5 - HISTORY Log

**Purpose:** Verify that the program displays all valid guesses and their feedback in the order they were entered.

**Input:**

WARNING: You have 6 guess(es) left.  
Please enter your guess: **GOYP**  
1B\_0W

WARNING: You have 5 guess(es) left.  
Please enter your guess: **HISTORY**

-----  
WARNING: You have 6 guess(es) left.  
Please enter your guess: **HAPPY**  
P A A A A

WARNING: You have 5 guess(es) left.  
Please enter your guess: **SHOUT**  
A P A A A

WARNING: You have 4 guess(es) left.  
Please enter your guess: **HISTORY**

**Expected Output:**

.  
. .  
. .  
. .  
GOYP: 1B\_0W

HAPPY: P A A A A  
SHOUT: A P A A A

**Actual Output:**

GBOY: 0B\_1W  
PRYY: 0B\_1W  
OOOO: 1B\_0W  
OBYP: 0B\_1W  
BOYP: 1B\_0W  
GOGR: 2B\_0W  
GOYP: 1B\_0W < Correct!

HAPPY: P A A A A  
SHOUT: A P A A A < Correct!

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| <b>Test 6 - Win Condition</b>                                                                                                                                                                                                                                                                                                           |
| <b>Purpose:</b> Verify that the program can correctly identify when the player guesses the secret <b>word</b> correctly.                                                                                                                                                                                                                |
| <b>Input:</b><br>Please enter your guess: <b>RBYP</b> (with Secret Code as: RBYP)<br>Please enter your guess: <b>BELOW</b> (with Secret Code as: BELOW)                                                                                                                                                                                 |
| <b>Expected Output:</b><br><br>4B_0W<br>You win!<br><br>Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)?<br><br>C C C C C<br>You win!<br><br>Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)?                                   |
| <b>Actual Output:</b><br><br>4B_0W<br>You win!<br><br>Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)? < <b>Correct!</b><br><br>C C C C C<br>You win!<br><br>Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)? < <b>Correct!</b> |

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| <b>Test 7 - Loss Condition</b>                                                                                                                                                                                                               |
| <b>Purpose:</b> Verify that the program can correctly identify when the player uses all their guesses without finding the secret <b>word</b> correctly.                                                                                      |
| <b>Input:</b><br><br>WARNING: You have 1 guess(es) left.<br>Please enter your guess: <b>GPOY</b> (with Secret Code as: GBRB)<br><br>WARNING: You have 1 guess(es) left.<br>Please enter your guess: <b>URNS</b> (with Secret Code as: BURNT) |
| <b>Expected Output:</b><br><br>1B_0W<br>Sorry, you lose! The secret code was GBRB.<br><br>Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)?                                               |

P C C C A

Sorry, you lose! The secret code was BURNT.

Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)?

**Actual Output:**

1B\_0W

Sorry, you lose! The secret code was GBRB.

Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)? < **Correct!**

P C C C A

Sorry, you lose! The secret code was BURNT.

Thank you for joining us for a game! Would you like to play another game (Please Enter 'Y' for Yes)? < **Correct!**

**Test 8 - Changed Configuration**

**Purpose:** Verify that the game algorithm works with a configuration different from the default by only changing the input in the 'Driver.java,' as there is a GameConfiguration class that should automatically update.

**Input:**

- Changed input in Driver.java: new **MastermindConfiguration**(3, 5, new char[]{'B', 'G', 'R'})

**Expected Output:**

Launching Mastermind Game [Test Mode: true]...

Guess the 3-peg secret code using the colors {BGR}. Colors can repeat.

NOTE: Type HISTORY at any time to see your past guesses.

[TEST MODE] Secret code generated for this game is: GBR

WARNING: You have 5 guess(es) left.

Please enter your guess:

**Actual Output:**

Launching Mastermind Game [Test Mode: true]...

Guess the 3-peg secret code using the colors {BGR}. Colors can repeat.

NOTE: Type HISTORY at any time to see your past guesses.

[TEST MODE] Secret code generated for this game is: GBR

WARNING: You have 5 guess(es) left.

Please enter your guess: < **Correct ('B' 'G' 'R' Colors + 3-peg + 5 guesses only!**

**Test 8 - Changed Configuration**

**Purpose:** Verify that the game algorithm works with a configuration different from the default by only changing the input in the 'Driver.java,' as there is a **WordleConfiguration** class that should automatically update.

**Input:**

- Changed input in Driver.java: `new WordleConfiguration(6, 6, "assignment2/Dictionary.txt")`

**Expected Output:**

Launching Wordle Game [Test Mode: true]...

Guess the 6-letter secret word. Your guess must also be a real word from the dictionary.

Feedback will be one symbol per letter: C = correct letter in the correct spot, P = letter is in the secret word but is in the wrong spot, A = letter is not in the secret word.

NOTE: Type HISTORY at any time to see your past guesses.

[TEST MODE] Secret code generated for this game is: CUBISM

WARNING: You have 6 guess(es) left.

Please enter your guess:

**Actual Output:**

Launching Wordle Game [Test Mode: true]...

Guess the 6-letter secret word. Your guess must also be a real word from the dictionary.

Feedback will be one symbol per letter: C = correct letter in the correct spot, P = letter is in the secret word but is in the wrong spot, A = letter is not in the secret word.

NOTE: Type HISTORY at any time to see your past guesses.

[TEST MODE] Secret code generated for this game is: CUBISM

WARNING: You have 6 guess(es) left.

Please enter your guess: < **Correct (6 Letters + 6 Guesses + Correct Dictionary File Loaded Successfully)!**